# Module 02: Agents — Generative Models, Sufficient Statistics, and the Free Energy Principle

## Learning Objectives

1. Define a **generative model** formally as a joint probability distribution $p(o, s, \theta)$ and distinguish it from the recognition density $q(s, \theta)$.
2. Derive the **sufficient statistics** of the recognition density and explain how they parameterize the agent's beliefs.
3. State the **Free Energy Principle** precisely: internal states parameterize a recognition density that minimizes free energy.

## Introduction

An agent, in the Active Inference framework, is a system whose internal states can be interpreted as parameterizing a recognition density — a probability distribution encoding beliefs about hidden states. This module formalizes what it means for a system to be an "agent" by developing the mathematics of generative models, sufficient statistics, and the Free Energy Principle.

## Key Concepts

### 1. The Generative Model

A generative model specifies how observations are generated from hidden causes:

$$\mathbf{p(o_{1:T}, s_{1:T}, \theta) = p(\theta) \prod_{t} p(o_{t} \mid s_{t}, \theta) \cdot p(s_{t} \mid s_{t-1}, \theta)}$$

where:

- $\mathbf{o_{1:T}}$ are observations (sensory data) over time
- $\mathbf{s_{1:T}}$ are hidden states (causes of observations)
- $\boldsymbol{\theta}$ are parameters (fixed properties of the environment)

- **$p(o_{?} \mid s_{?}, \theta)$** is the likelihood (how states generate observations)
- **$p(s_{?} \mid s_{?-1}, \theta)$** are transition probabilities (how states evolve)
- **$p(\theta)$** is the prior over parameters

This is a partially observed Markov decision process (POMDP) when extended with actions.

## 2. The Recognition Density and Sufficient Statistics

Since exact Bayesian inference $p(s, \theta \mid o)$ is intractable for most models, the agent maintains an approximate posterior — the **recognition density** $q(s, \theta)$:

$$q(s, \theta) \approx p(s, \theta \mid o)$$

Under the mean-field approximation, this factorizes:

$$q(s, \theta) = q(s) \cdot q(\theta)$$

For exponential family distributions, q is fully specified by its **sufficient statistics**. For example, a Gaussian $q(s) = N(s; \mu, \Sigma)$ is parameterized by mean $\mu$ and covariance $\Sigma$. These sufficient statistics are the **internal states** of the agent — the quantities that the brain (or any self-organizing system) must track.

## 3. The Free Energy Principle (Formal Statement)

**Free Energy Principle**: For any system with a Markov Blanket at nonequilibrium steady state, the internal states $\mu$ can be described as parameterizing a recognition density $q_\mu(\eta)$ whose variational free energy is minimized:

$$\mu^* = \text{argmin}_\mu F(s, \mu) = \text{argmin}_\mu E_{q\mu}[\ln q_\mu(\eta) - \ln p(\eta, s)]$$

This is not a claim that all systems *compute* free energy. It is that the dynamics of any Markov-blanketed system can be *described as if* they minimize free energy — a formal duality between physics and inference.

## Derivation Exercises

1. Write the generative model for a simple Hidden Markov Model with K discrete states and D-dimensional Gaussian observations. Identify all parameters.
2. For a Gaussian recognition density $q(s) = N(\mu, \sigma^2)$, compute the free energy $F = E_q[\ln q(s)] - E_q[\ln p(o, s)]$ for a generative model $p(s) = N(0, 1)$, $p(o|s) = N(s, \sigma_o^2)$.
3. Derive the update equations for the sufficient statistics $\mu$ and $\sigma^2$ by setting $\partial F/\partial \mu = 0$ and $\partial F/\partial \sigma^2 = 0$.

## Conclusion

The agent is defined by its generative model and its recognition density. The Free Energy Principle establishes that any self-organizing system can be interpreted as performing approximate Bayesian inference through free energy minimization. Module 03 extends this to perception — updating beliefs about hidden states given new observations.